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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 31st January Evening Shift

JEE Main Question Paper with Answer Key

31 Jan 2023	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

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Physics	28 questions
Out of syllabus	18 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

When a hydrocarbon A undergoes complete combustion it requires 11 equivalents of oxygen and produces 4 equivalents of water. What is the molecular formula of A ?

(A) C_9H_8 (B) C_5H_8 (C) $C_{11}H_4$ (D) $C_{11}H_8$

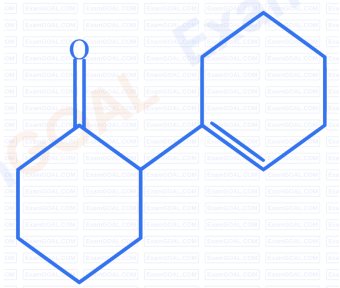
Open solution

Q2. MCQ (Single Correct Answer)

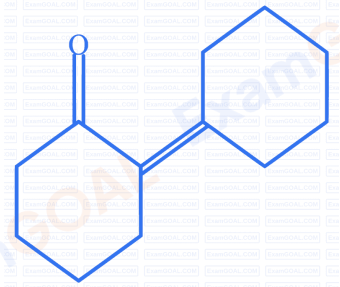
Cyclohexylamine when treated with nitrous acid yields (P). On treating (P) with PCC results in (Q). When (Q) is heated with dil. $NaOH$ we get (R) The final product (R) is :

Open solution

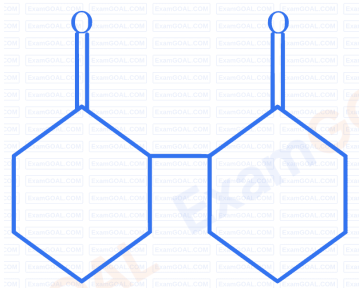
(A)



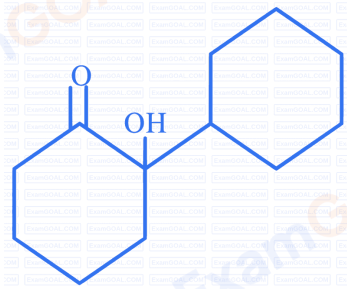
(B)



(C)



(D)

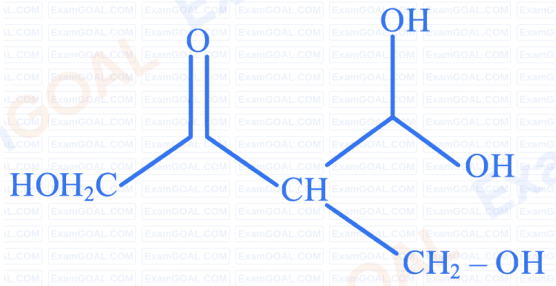


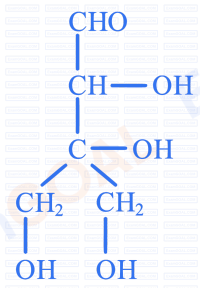
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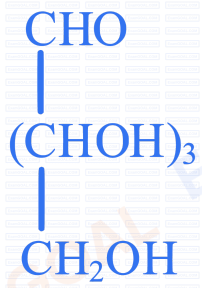
Q3. MCQ (Single Correct Answer)

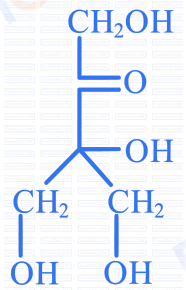
Open solution

Compound $A, C_5H_{10}O_5$, gives a tetraacetate with Ac_2O and oxidation of A with $Br_2 - H_2O$ gives an acid, $C_5H_{10}O_6$. Reduction of A with HI gives isopentane. The possible structure of A is

(A) 

(B) 

(C) 

(D) 

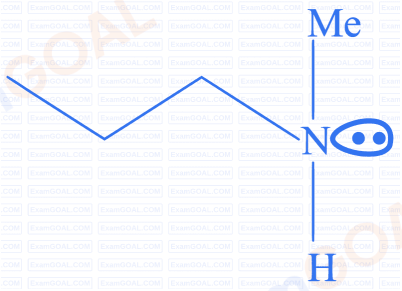
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Q4. MCQ (Single Correct Answer)

Open solution

An organic compound $[A]$ $(C_4H_{11}N)$, shows optical activity and gives N_2 gas on treatment with HNO_2. The compound $[A]$ reacts with $PhSO_2Cl$ producing a compound which is soluble in KOH. The structure of A is :

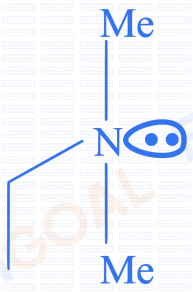
(A)



(B)



(C)



(D)



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Q5. MCQ (Single Correct Answer)

Open solution

Which of the following elements have half-filled f-orbitals in their ground state?

(Given : atomic number $\mathrm{Sm}=62; \mathrm{Eu}=63; \mathrm{Tb}=65; \mathrm{Gd}=64, \mathrm{Pm}=61$)

A. Sm

B. Eu

C. Tb

D. Gd

E. Pm

Choose the correct answer from the options given below :

(A) B and D only **(B)** A and B only **(C)** C and D only **(D)** A and E only

Q6. MCQ (Single Correct Answer)

Open solution

Evaluate the following statements for their correctness.

A. The elevation in boiling point temperature of water will be same for $0.1\mathrm{M}\,\mathrm{NaCl}$ and $0.1\mathrm{M}$ urea.

B. Azeotropic mixtures boil without change in their composition.

C. Osmosis always takes place from hypertonic to hypotonic solution.

D. The density of $32\%\,\mathrm{H_2SO_4}$ solution having molarity $4.09\,\mathrm{M}$ is approximately $1.26\,\mathrm{g\,mL^{-1}}$

E. A negatively charged sol is obtained when KI solution is added to silver nitrate solution.

Choose the correct answer from the options given below :

(A) A, B and D only **(B)** A and C only **(C)** B and D only **(D)** B, D and E only

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Q7. MCQ (Single Correct Answer)

Open solution

Arrange the following orbitals in decreasing order of energy.

A. $n = 3, l = 0, m = 0$

B. $n = 4, l = 0, m = 0$

C. $n = 3, l = 1, m = 0$

D. $n = 3, l = 2, m = 1$

The correct option for the order is :

(A) $B > D > C > A$

(B) $A > C > B > D$

(C) $D > B > A > C$

(D) $D > B > C > A$

Q8. MCQ (Single Correct Answer)

Open solution

In Dumas method for the estimation of N_2, the sample is heated with copper oxide and the gas evolved is passed over :

(A) Copper oxide

(B) Copper gauze

(C) Pd

(D) Ni

Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): The first ionization enthalpy of $3d$ series elements is more than that of group 2 metals

Reason (R): In 3d series of elements successive filling of d-orbitals takes place.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) (A) is true but (R) is false

(C) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(D) (A) is false but (R) is true

Q10. MCQ (Single Correct Answer)

Open solution

A hydrocarbon ' X ' with formula C_6H_8 uses two moles of H_2 on catalytic hydrogenation of its one mole. On ozonolysis, ' X ' yields two moles of methane dicarbaldehyde. The hydrocarbon ' X ' is:

(A) hexa-1, 3, 5-triene

(B) cyclohexa-1, 4-diene

(C) cyclohexa - 1,3 - diene

(D) 1-methylcyclopenta-1, 4-diene

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Q11. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I: Upon heating a borax bead dipped in cupric sulphate in a luminous flame, the colour of the bead becomes green

Statement II: The green colour observed is due to the formation of copper(I) metaborate

In light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q12. MCQ (Single Correct Answer)

Open solution

In the following halogenated organic compounds, the one with the maximum number of chlorine atoms in its structure is :

(A) Gammaxene (B) Chloropicrin (C) Chloral (D) Freon-12

Q13. MCQ (Single Correct Answer)

Open solution

The incorrect statement for the use of indicators in acid-base titration is :

(A) Phenolphthalein is a suitable indicator for a weak acid vs strong base titration.

(B) Methyl orange may be used for a weak acid vs weak base titration.

(C) Methyl orange is a suitable indicator for a strong acid vs weak base titration.

(D) Phenolphthalein may be used for a strong acid vs strong base titration.

Q14. Numerical

Open solution

Assume carbon burns according to the following equation :

$$2C_{(s)} + O_{2(g)} \rightarrow 2CO(g)$$

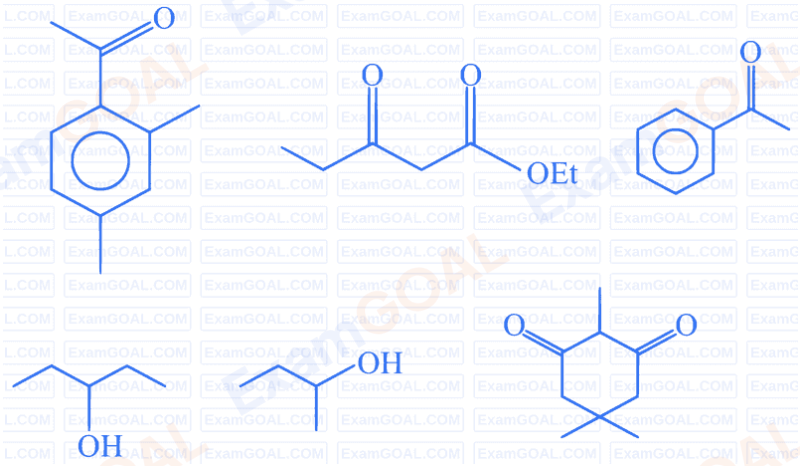
when $12\ g$ carbon is burnt in $48\ g$ of oxygen, the volume of carbon monoxide produced is _____ $\times 10^{-1}\ L$ at STP [nearest integer]

[Given: Assume CO as ideal gas, Mass of C is $12\ g\ mol^{-1}$, Mass of O is $16\ g\ mol^{-1}$ and molar volume of an ideal gas at STP is $22.7\ L\ mol^{-1}$]

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Q15. Numerical

The number of molecules which gives the haloform test among the following molecules is _____.



Open solution

Q16. Numerical

Amongst the following, the number of species having the linear shape is _____.

$XeF_2, I_3^+, C_3O_2, I_3^-, CO_2, SO_2, BeCl_2$ and $BCl_2^{\ominus}$

Open solution

Q17. Numerical

Enthalpies of formation of $CCl_4(g), H_2O(g), CO_2(g)$ and $HCl(g)$ are $-105, -242, -394$ and $-92\ kJ\ mol^{-1}$ respectively. The magnitude of enthalpy of the reaction given below is _____ $kJ\ mol^{-1}$. (nearest integer)

$$CCl_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4HCl(g)$$

Open solution

Q18. Numerical

At $298\ K$, the solubility of silver chloride in water is $1.434 \times 10^{-3}\ g\ L^{-1}$. The value of $-\log K_{sp}$ for silver chloride is _____.

(Given mass of Ag is $107.9\ g\ mol^{-1}$ and mass of Cl is $35.5\ g\ mol^{-1}$)

Open solution

Q19. Numerical

The resistivity of a $0.8M$ solution of an electrolyte is $5 \times 10^{-3} \Omega\ cm$.

Its molar conductivity is _____ $\times 10^4\ \Omega^{-1}\ cm^2\ mol^{-1}$. (Nearest integer)

Open solution

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Q20. Numerical

A sample of a metal oxide has formula $M_{0.83}O_{1.00}$. The metal M can exist in two oxidation states $+2$ and $+3$.

In the sample of $M_{0.83}O_{1.00}$, the percentage of metal ions existing in $+2$ oxidation state is _____ %. (nearest integer)

Open solution

Q21. Numerical

If the CFSE of $[Ti(H_2O)_6]^{3+}$ is $-96.0\ kJ/mol$, this complex will absorb maximum at wavelength _____ nm. (nearest integer)

Assume Planck's constant (h) $= 6.4 \times 10^{-34} Js$, Speed of light $(c) = 3.0 \times 10^8\ m/s$ and Avogadro's

Constant $(N_A) = 6 \times 10^{23}/mol$

Open solution

Q22. Numerical

The rate constant for a first order reaction is $20\ min^{-1}$. The time required for the initial concentration of the reactant to reduce to its $\frac{1}{32}$ level is _____ $\times 10^{-2}$ min. (Nearest integer)

(Given : $\ln 10 = 2.303$ and $\log 2 = 0.3010$)

Open solution

Mathematics 22 questions

Q23. MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$ be three vectors. If $\vec{r}$ is a vector such that, $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then $25|\vec{r}|^2$ is equal to :

(A) 336 (B) 449 (C) 339 (D) 560

Open solution

Q24. MCQ (Single Correct Answer)

Let $f : \mathbb{R} - \{2, 6\} \to \mathbb{R}$ be real valued function

defined as $f(x) = \frac{x^2+2x+1}{x^2-8x+12}$.

Then range of f is

(A) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

(B) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(C) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(D) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

Open solution

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Q25. MCQ (Single Correct Answer)

The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in \mathbb{R}$ has :

(A) two solutions and both are negative

(B) two solutions and only one of them is negative

(C) four solutions two of which are negative

(D) no solution

Open solution

Q26. MCQ (Single Correct Answer)

Let (a, b) $\subset (0, 2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta) - \cos^{-1}(\sin\theta) > 0, \theta \in (0, 2\pi)$, holds.

If $\alpha x^2 + \beta x + \sin^{-1}\left(x^2 - 6x + 10\right) + \cos^{-1}\left(x^2 - 6x + 10\right) = 0$ and $\alpha - \beta = b - a$, then α is equal to :

(A) $\frac{\pi}{16}$ (B) $\frac{\pi}{48}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{12}$

Open solution

Q27. MCQ (Single Correct Answer)

Let $\alpha > 0$. If $\int_0^{\alpha} \frac{x}{\sqrt{x+\alpha}-\sqrt{x}}\,dx = \frac{16+20\sqrt{2}}{15}$, then α is equal to :

(A) 4 (B) 2 (C) $2\sqrt{2}$ (D) $\sqrt{2}$

Open solution

Q28. MCQ (Single Correct Answer)

Let the mean and standard deviation of marks of class A of 100 students be respectively 40 and $\alpha(>0)$, and the mean and standard deviation of marks of class B of n students be respectively 55 and $30-\alpha$. If the mean and variance of the marks of the combined class of $100+\mathrm{n}$ students are respectively 50 and 350 , then the sum of variances of classes A and B is :

(A) 450 (B) 900 (C) 650 (D) 500

Open solution

Q29. MCQ (Single Correct Answer)

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $a_7 = 3$, the product $a_1 a_4$ is minimum and the sum of its first n terms is zero, then $n! - 4a_{n(n+2)}$ is equal to :

(A) 24 (B) $\frac{381}{4}$ (C) 9 (D) $\frac{33}{4}$

Open solution

Q30. MCQ (Single Correct Answer)

The complex number $z = \frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$ is equal to :

(A) $\cos\frac{\pi}{12} - i\sin\frac{\pi}{12}$ (B) $\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

(C) $\sqrt{2}i\left(\cos\frac{5\pi}{12} - i\sin\frac{5\pi}{12}\right)$ (D) $\sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$

Open solution

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Q31. MCQ (Single Correct Answer)

Open solution

Among the relations

$S = \{(a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0\}$

and $T = \{(a, b) : a, b \in \mathbb{R}, a^2 - b^2 \in \mathbb{Z}\}$,

(A) S is transitive but T is not

(B) both S and T are symmetric

(C) neither S nor T is transitive

(D) T is symmetric but S is not

Q32. MCQ (Single Correct Answer)

Open solution

The absolute minimum value, of the function

$f(x) = |x^2 - x + 1| + [x^2 - x + 1]$,

where $[t]$ denotes the greatest integer function, in the interval $[-1, 2]$, is :

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$ (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Q33. MCQ (Single Correct Answer)

Open solution

If $\phi(x) = \frac{1}{\sqrt{x}} \int_{\frac{\pi}{4}}^{x} \left(4\sqrt{2}\sin t - 3\phi'(t)\right) dt, x > 0,$

then $\phi'\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{4}{6+\sqrt{\pi}}$ (B) $\frac{4}{6-\sqrt{\pi}}$ (C) $\frac{8}{\sqrt{\pi}}$ (D) $\frac{8}{6+\sqrt{\pi}}$

Q34. MCQ (Single Correct Answer)

Open solution

Let H be the hyperbola, whose foci are $(1 \pm \sqrt{2}, 0)$ and eccentricity is $\sqrt{2}$. Then the length of its latus rectum is :

(A) $\frac{5}{2}$ (B) 3 (C) 2 (D) $\frac{3}{2}$

Q35. MCQ (Single Correct Answer)

Open solution

$\lim\limits_{x\to\infty} \frac{(\sqrt{3x+1}+\sqrt{3x-1})^6+(\sqrt{3x+1}-\sqrt{3x-1})^6}{\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6} x^3$

(A) is equal to 9

(B) is equal to $\frac{27}{2}$

(C) does not exist

(D) is equal to 27

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Q36. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation

$$\left(3y^2 - 5x^2\right)y\,dx + 2x\left(x^2 - y^2\right)dy = 0$$

such that $y(1) = 1$. Then $\left|(y(2))^3 - 12y(2)\right|$ is equal to :

(A) 64 (B) $16\sqrt{2}$ (C) 32 (D) $32\sqrt{2}$

Open solution

Q37. MCQ (Single Correct Answer)

The set of all values of a^2 for which the line $x + y = 0$ bisects two distinct chords drawn from a point $\mathrm{P}\left(\frac{1+a}{2}, \frac{1-a}{2}\right)$ on the circle $2x^2 + 2y^2 - (1+a)x - (1-a)y = 0$, is equal to :

(A) $(0, 4]$ (B) $(4, \infty)$ (C) $(2, 12]$ (D) $(8, \infty)$

Open solution

Q38. Numerical

Let the area of the region

$\left\{(x, y) : |2x - 1| \le y \le \left|x^2 - x\right|, 0 \le x \le 1\right\}$ be A.

Then $(6\,\mathrm{A} + 11)^2$ is equal to

Open solution

Q39. Numerical

The coefficient of x^{-6}, in the

expansion of $\left(\frac{4x}{5} + \frac{5}{2x^2}\right)^9$, is

Open solution

Q40. Numerical

Let $\mathrm{A} = [\mathrm{a}_{ij}], \mathrm{a}_{ij} \in \mathbb{Z} \cap [0, 4], 1 \le i, j \le 2$.

The number of matrices A such that the sum of all entries is a prime number $\mathrm{p} \in (2, 13)$ is _____.

Open solution

Q41. Numerical

If ${}^{2n+1}\mathrm{P}_{n-1} : {}^{2n-1}\mathrm{P}_n = 11 : 21$,

then $n^2 + n + 15$ is equal to :

Open solution

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Q42. Numerical

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that

$|\vec{a}| = \sqrt{31}, 4|\vec{b}| = |\vec{c}| = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$.

If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ is equal to _____.

Open solution

Q43. Numerical

Let A be the event that the absolute difference between two randomly chosen real numbers in the sample space $[0, 60]$ is less than or equal to a . If $\mathrm{P(A)} = \frac{11}{36}$, then a is equal to _____.

Open solution

Q44. Numerical

If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2} - \frac{4}{x^l}\right)^9$ is -84 and the coefficient of x^{-3l} is $2^\alpha \beta$, where $\beta < 0$ is an odd number, then $|\alpha l - \beta|$ is equal to _____.

Open solution

Physics 28 questions

Q45. MCQ (Single Correct Answer)

For a solid rod, the Young's modulus of elasticity is $3.2 \times 10^{11} \mathrm{Nm}^{-2}$ and density is $8 \times 10^3\ \mathrm{kg\,m}^{-3}$. The velocity of longitudinal wave in the rod will be.

(A) $3.65 \times 10^3\ \mathrm{ms}^{-1}$ (B) $6.32 \times 10^3\ \mathrm{ms}^{-1}$

(C) $18.96 \times 10^3\ \mathrm{ms}^{-1}$ (D) $145.75 \times 10^3\ \mathrm{ms}^{-1}$

Open solution

Q46. MCQ (Single Correct Answer)

Heat energy of $735\ \mathrm{J}$ is given to a diatomic gas allowing the gas to expand at constant pressure. Each gas molecule rotates around an internal axis but do not oscillate. The increase in the internal energy of the gas will be :

(A) $572\ \mathrm{J}$ (B) $441\ \mathrm{J}$ (C) $525\ \mathrm{J}$ (D) $735\ \mathrm{J}$

Open solution

Q47. MCQ (Single Correct Answer)

The radius of electron's second stationary orbit in Bohr's atom is R. The radius of 3rd orbit will be

(A) 2.25R (B) $3\mathrm{R}$ (C) $\frac{\mathrm{R}}{3}$ (D) $9\mathrm{R}$

Open solution

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Q48. MCQ (Single Correct Answer)

If the two metals A and B are exposed to radiation of wavelength $350\ \mathrm{nm}$. The work functions of metals A and B are $4.8\mathrm{eV}$ and $2.2\mathrm{eV}$. Then choose the correct option.

(A) Metal B will not emit photo-electrons

(B) Both metals A and B will not emit photo-electrons

(C) Metal A will not emit photo-electrons

(D) Both metals A and B will emit photo-electrons

Open solution

Q49. MCQ (Single Correct Answer)

The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be :

(A) 0% (B) 75% (C) 100% (D) 50%

Open solution

Q50. MCQ (Single Correct Answer)

Match List **I** with List **II**

LIST I		**LIST II**	
A.	Microwaves	I.	Physiotherapy
B.	UV rays	II.	Treatment of cancer
C.	Infra-red light	III.	Lasik eye surgery
D.	X-ray	IV.	Aircraft navigation

Choose the correct answer from the options given below:

(A) A - IV, B - III, C - I, D - II

(B) A - II, B - IV, C - III, D - I

(C) A - III, B - II, C - I, D - IV

(D) A - IV, B - I, C - II, D - III

Open solution

Q51. MCQ (Single Correct Answer)

The H amount of thermal energy is developed by a resistor in $10\ \mathrm{s}$ when a current of $4\ \mathrm{A}$ is passed through it. If the current is increased to $16\ \mathrm{A}$, the thermal energy developed by the resistor in $10\ \mathrm{s}$ will be :

(A) $\frac{\mathrm{H}}{4}$ (B) $16\mathrm{H}$ (C) H (D) $4\mathrm{H}$

Open solution

Q52. MCQ (Single Correct Answer)

A body weight W, is projected vertically upwards from earth's surface to reach a height above the earth which is equal to nine times the radius of earth. The weight of the body at that height will be :

(A) $\frac{W}{91}$ (B) $\frac{\mathrm{W}}{3}$ (C) $\frac{\mathrm{W}}{100}$ (D) $\frac{\mathrm{W}}{9}$

Open solution

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Q53. MCQ (Single Correct Answer)

Considering a group of positive charges, which of the following statements is correct ?

(A) Net potential of the system cannot be zero at a point but net electric field can be zero at that point

(B) Net potential of the system at a point can be zero but net electric field can't be zero at that point.

(C) Both the net potential and the net electric field cannot be zero at a point.

(D) Both the net potential and the net field can be zero at a point.

Open solution

Q54. MCQ (Single Correct Answer)

A body of mass $10\ \mathrm{kg}$ is moving with an initial speed of $20\ \mathrm{m/s}$. The body stops after $5\ \mathrm{s}$ due to friction between body and the floor. The value of the coefficient of friction is:

(Take acceleration due to gravity $g = 10\ \mathrm{ms}^{-2}$)

(A) 0.3 (B) 0.2 (C) 0.5 (D) 0.4

Open solution

Q55. MCQ (Single Correct Answer)

A hypothetical gas expands adiabatically such that its volume changes from 8 litres to 27 litres. If the ratio of final pressure of the gas to initial pressure of the gas is $\frac{16}{81}$. Then the ratio of $\frac{\mathrm{Cp}}{\mathrm{Cv}}$ will be.

(A) $\frac{3}{1}$ (B) $\frac{4}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Open solution

Q56. MCQ (Single Correct Answer)

A long conducting wire having a current I flowing through it, is bent into a circular coil of N turns. Then it is bent into a circular coil of n turns. The magnetic field is calculated at the centre of coils in both the cases. The ratio of the magnetic field in first case to that of second case is :

(A) $N^2 : n^2$ (B) $\mathrm{N} : \mathrm{n}$ (C) $\mathrm{n} : \mathrm{N}$ (D) $n^2 : N^2$

Open solution

Q57. MCQ (Single Correct Answer)

A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index $\frac{5}{3}$ is poured inside the bucket, then the microscope has to be raised by $30\ \mathrm{cm}$ to focus the object again. The height of the liquid in the bucket is :

(A) $50\ \mathrm{cm}$ (B) $18\ \mathrm{cm}$ (C) $75\ \mathrm{cm}$ (D) $12\ \mathrm{cm}$

Open solution

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Q58. MCQ (Single Correct Answer)

Match List **I** with List **II**

LIST I		LIST II	
A.	Angular momentum	I.	$\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$
B.	Torque	II.	$\left[\mathrm{ML}^{-2}\,\mathrm{T}^{-2}\right]$
C.	Stress	III	$\left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]$
D.	Pressure gradient	IV.	$\left[\mathrm{ML}^{-1}\,\mathrm{T}^{-2}\right]$

Open solution

Choose the correct answer from the options given below:

(A) A - I, B - IV, C - III, D - II

(B) A - III, B - I, C - IV, D - II

(C) A - IV, B - II, C - I, D - III

(D) A - II, B - III, C - IV, D - I

Q59. MCQ (Single Correct Answer)

Under the same load, wire A having length $5.0\,\mathrm{m}$ and cross section $2.5 \times 10^{-5}\,\mathrm{m}^2$ stretches uniformly by the same amount as another wire B of length $6.0\,\mathrm{m}$ and a cross section of $3.0 \times 10^{-5}\,\mathrm{m}^2$ stretches. The ratio of the Young's modulus of wire A to that of wire B will be :

Open solution

(A) $1:2$

(B) $1:4$

(C) $1:1$

(D) $1:10$

Q60. MCQ (Single Correct Answer)

An alternating voltage source $\mathrm{V} = 260\sin(628\mathrm{t})$ is connected across a pure inductor of $5\mathrm{mH}$. Inductive reactance in the circuit is :

Open solution

(A) 6.28Ω

(B) 0.318Ω

(C) 0.5Ω

(D) 3.14Ω

Q61. MCQ (Single Correct Answer)

A stone of mass $1\,\mathrm{kg}$ is tied to end of a massless string of length $1\,\mathrm{m}$. If the breaking tension of the string is $400\,\mathrm{N}$, then maximum linear velocity, the stone can have without breaking the string, while rotating in horizontal plane, is :

Open solution

(A) $20\,\mathrm{ms}^{-1}$

(B) $40\,\mathrm{ms}^{-1}$

(C) $400\,\mathrm{ms}^{-1}$

(D) $10\,\mathrm{ms}^{-1}$

Q62. MCQ (Single Correct Answer)

A body is moving with constant speed, in a circle of radius $10\,\mathrm{m}$. The body completes one revolution in $4\,\mathrm{s}$. At the end of 3rd second, the displacement of body (in m) from its starting point is :

Open solution

(A) 15π

(B) 30

(C) $10\sqrt{2}$

(D) 5π

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Q63. Numerical

A water heater of power $2000\ \mathrm{W}$ is used to heat water. The specific heat capacity of water is $4200\ \mathrm{J\ kg^{-1}\ K^{-1}}$. The efficiency of heater is 70%. Time required to heat $2\ \mathrm{kg}$ of water from $10^{\circ}\mathrm{C}$ to $60^{\circ}\mathrm{C}$ is _____ s.

(Assume that the specific heat capacity of water remains constant over the temperature range of the water).

Open solution

Q64. Numerical

A series LCR circuit consists of $\mathrm{R} = 80\Omega$, $\mathrm{X_L} = 100\Omega$, and $\mathrm{X_C} = 40\Omega$. The input voltage is $2500\cos(100\pi t)\mathrm{V}$. The amplitude of current, in the circuit, is _____ A.

Open solution

Q65. Numerical

Two light waves of wavelengths 800 and $600\ \mathrm{nm}$ are used in Young's double slit experiment to obtain interference fringes on a screen placed $7\ \mathrm{m}$ away from plane of slits. If the two slits are separated by $0.35\ \mathrm{mm}$, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be _____ mm.

Open solution

Q66. Numerical

Two bodies are projected from ground with same speeds $40\ \mathrm{ms^{-1}}$ at two different angles with respect to horizontal. The bodies were found to have same range. If one of the body was projected at an angle of 60°, with horizontal then sum of the maximum heights, attained by the two projectiles, is _____ m. (Given $\mathrm{g} = 10\ \mathrm{ms^{-2}}$)

Open solution

Q67. Numerical

The displacement equations of two interfering waves are given by

$y_1 = 10\sin\left(\omega t + \frac{\pi}{3}\right)\mathrm{cm}$, $y_2 = 5[\sin\omega t + \sqrt{3}\cos\omega t]\mathrm{cm}$ respectively.

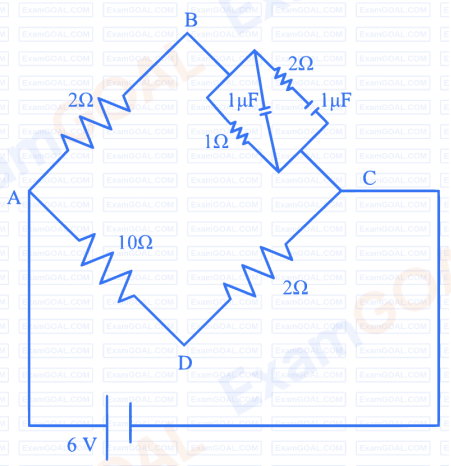
The amplitude of the resultant wave is _____ cm.

Open solution

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Q68. Numerical

For the given circuit, in the steady state, $|V_B - V_D| =$ _____ V.



Open solution

Q69. Numerical

Two discs of same mass and different radii are made of different materials such that their thicknesses are $1\ \text{cm}$ and $0.5\ \text{cm}$ respectively. The densities of materials are in the ratio $3:5$. The moment of inertia of these discs respectively about their diameters will be in the ratio of $\frac{x}{6}$. The value of x is _____.

Open solution

Q70. Numerical

A ball is dropped from a height of $20\ \text{m}$. If the coefficient of restitution for the collision between ball and floor is 0.5, after hitting the floor, the ball rebounds to a height of _____ m.

Open solution

Q71. Numerical

If the binding energy of ground state electron in a hydrogen atom is $13.6\ \text{eV}$, then, the energy required to remove the electron from the second excited state of Li^{2+} will be : $x \times 10^{-1}\text{eV}$. The value of x is _____.

Open solution

Q72. Numerical

Two parallel plate capacitors C_1 and C_2 each having capacitance of $10\mu\text{F}$ are individually charged by a 100 V D.C. source. Capacitor C_1 is kept connected to the source and a dielectric slab is inserted between its plates. Capacitor C_2 is disconnected from the source and then a dielectric slab is inserted in it. Afterwards the capacitor C_1 is also disconnected from the source and the two capacitors are finally connected in parallel combination. The common potential of the combination will be _____ V.

(Assuming Dielectric constant $= 10$)

Open solution

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Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q73. MCQ (Single Correct Answer) · **Chemistry**

The normal rain water is slightly acidic and its pH value is 5.6 because of which one of the following?

Open solution

(A) $N_2O_5 + H_2O \rightarrow 2HNO_3$

(B) $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

(C) $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$

(D) $CO_2 + H_2O \rightarrow H_2CO_3$

Q74. MCQ (Single Correct Answer) · **Chemistry**

The element playing significant role in neuromuscular function and interneuronal transmission is :

Open solution

(A) Be (B) Ca (C) Mg (D) Li

Q75. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : H_2O_2 is used in the synthesis of Cephalosporin

Statement II : H_2O_2 is used for the restoration of aerobic conditions to sewage wastes.

Open solution

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement $\mathbf{I}$ is incorrect but Statement $\mathbf{II}$ is correct

(B) Statement $\mathbf{I}$ is correct but Statement $\mathbf{II}$ is incorrect

(C) Both Statement $\mathbf{I}$ and Statement $\mathbf{II}$ are incorrect

(D) Both Statement $\mathbf{I}$ and Statement $\mathbf{II}$ are correct

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List **I** with List **II**

LIST I		**LIST II**	
A.	Physisorption	I.	Single Layer Adsorption
B.	Chemisorption	II.	$20-40\ \mathrm{kJ\ mol^{-1}}$
C.	$\mathrm{N_2(g)+3H_2(g)\xrightarrow{Fe(s)}2NH_3(g)}$	III.	Chromatography
D.	Analytical Application or Adsorption	IV.	Heterogeneous catalysis

Choose the **correct** answer from the options given below:

(A) A - III, B - IV, C - I, D - II

(B) A - II, B - III, C - I, D - IV

(C) A - II, B - I, C - IV, D - III

(D) A - IV, B - II, C - III, D - I

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following compounds are not used as disinfectants ?

A. Chloroxylenol

B. Bithional

C. Veronal

D. Prontosil

E. Terpineol

Choose the correct answer from the options given below :

(A) A, B

(B) $\mathrm{C, D}$

(C) B, D, E

(D) $\mathrm{A, B, E}$

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The Lewis acid character of boron tri halides follows the order :

(A) $\mathrm{BI_3 > BBr_3 > BCl_3 > BF_3}$

(B) $\mathrm{BCl_3 > BF_3 > BBr_3 > BI_3}$

(C) $\mathrm{BF_3 > BCl_3 > BBr_3 > BI_3}$

(D) $\mathrm{BBr_3 > BI_3 > BCl_3 > BF_3}$

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Q79. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following statements is incorrect?

(A) Cast iron is obtained by melting pig iron with scrap iron and coke using hot air blast.

(B) The malleable iron is prepared from cast iron by oxidising impurities in a reverberatory furnace.

(C) van Arkel method is used to purify tungsten.

(D) Boron and Indium can be purified by zone refining method.

Open solution

Q80. Numerical · **Chemistry**

The number of alkali metal(s), from $\mathrm{Li}, \mathrm{K}, \mathrm{Cs}$, Rb having ionization enthalpy greater than $400\ \mathrm{kJ\ mol^{-1}}$ and forming stable super oxide is _____.

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let the plane $\mathrm{P}: 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to the line $\mathrm{L}: \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P on the y-axis is 1 , then the distance between P and L is :

(A) $\frac{6}{\sqrt{14}}$ (B) $\sqrt{14}$ (C) $\sqrt{\frac{2}{7}}$ (D) $\sqrt{\frac{7}{2}}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The number of values of $\mathrm{r} \in \{\mathrm{p}, \mathrm{q}, \sim \mathrm{p}, \sim \mathrm{q}\}$ for which $((\mathrm{p} \wedge \mathrm{q}) \Rightarrow (\mathrm{r} \vee \mathrm{q})) \wedge ((\mathrm{p} \wedge \mathrm{r}) \Rightarrow \mathrm{q})$ is a tautology, is :

(A) 2 (B) 1 (C) 4 (D) 3

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points $\mathrm{A}, \mathrm{B}, \mathrm{C}$ is $(2, \mathrm{a}, 4), \mathrm{a} \in \mathrm{N}$. If the volume of the tetrahedron OABC is 144 unit^3, then which of the following points is **NOT** on P ?

(A) $(3, 0, 4)$ (B) $(0, 6, 3)$ (C) $(0, 4, 4)$ (D) $(2, 2, 4)$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let P be the plane, passing through the point $(1, -1, -5)$ and perpendicular to the line joining the points $(4, 1, -3)$ and $(2, 4, 3)$. Then the distance of P from the point $(3, -2, 2)$ is :

(A) 5 (B) 4 (C) 6 (D) 7

Open solution

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Q85. MCQ (Single Correct Answer) · **Mathematics**

Open solution

If a point $\mathrm{P}(\alpha, \beta, \gamma)$ satisfying

$$(\alpha \quad \beta \quad \gamma)\begin{pmatrix}2 & 10 & 8\\ 9 & 3 & 8\\ 8 & 4 & 8\end{pmatrix} = (0 \quad 0 \quad 0)$$

lies on the plane $2x + 4y + 3z = 5$, then $6\alpha + 9\beta + 7\gamma$ is equal to :

(A) $\frac{11}{5}$ (B) 11 (C) -1 (D) $\frac{5}{4}$

Q86. Numerical · **Mathematics**

Open solution

Let A be a $n \times n$ matrix such that $|\mathrm{A}| = 2$. If the determinant of the matrix $\mathrm{Adj}\left(2 \cdot \mathrm{Adj}\left(2\,\mathrm{A}^{-1}\right)\right)$ is 2^{84}, then n is equal to :

Q87. Numerical · **Mathematics**

Open solution

Let S be the set of all $\mathrm{a} \in \mathrm{N}$ such that the area of the triangle formed by the tangent at the point $\mathrm{P}(\mathrm{b}$, c), b, c $\in \mathbb{N}$, on the parabola $y^2 = 2\mathrm{a}x$ and the lines $x = \mathrm{b}, y = 0$ is 16 unit^2, then $\sum\limits_{\mathrm{a}\in\mathrm{S}} \mathrm{a}$ is equal to :

Q88. Numerical · **Mathematics**

Open solution

The sum $1^2 - 2 \cdot 3^2 + 3 \cdot 5^2 - 4 \cdot 7^2 + 5 \cdot 9^2 - \ldots + 15 \cdot 29^2$ is _____.

Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements :

Statement I : For transmitting a signal, size of antenna (l) should be comparable to wavelength of signal (at least $l = \frac{\lambda}{4}$ in dimension)

Statement II: In amplitude modulation, amplitude of carrier wave remains constant (unchanged).

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

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Q90. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements :

Statement I: In a typical transistor, all three regions emitter, base and collector have same doping level.

Statement II: In a transistor, collector is the thickest and base is the thinnest segment.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) **Statement I** is correct but **Statement II** is incorrect

(B) Both **Statement I** and **Statement II** are incorrect

(C) **Statement I** is incorrect but **Statement II** is correct

(D) Both **Statement I** and **Statement II** are correct

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Answer Key

Chemistry

1. **A**	2. **B**	3. **B**	4. **B**	5. **A**
6. **C**	7. **D**	8. **B**	9. **A**	10. **B**
11. **A**	12. **A**	13. **B**	14. **227**	15. **3**
16. **5**	17. **173**	18. **10**	19. **25**	20. **59**
21. **480**	22. **17**			

Mathematics

23. **C**	24. **C**	25. **A**	26. **D**	27. **B**
28. **D**	29. **A**	30. **D**	31. **D**	32. **A**
33. **D**	34. **C**	35. **D**	36. **D**	37. **D**
38. **125**	39. **5040**	40. **204**	41. **45**	42. **3**
43. **10**	44. **98**			

Physics

45. **B**	46. **C**	47. **A**	48. **C**	49. **A**
50. **A**	51. **B**	52. **C**	53. **A**	54. **D**
55. **B**	56. **A**	57. **C**	58. **B**	59. **C**
60. **D**	61. **A**	62. **C**	63. **300**	64. **25**
65. **48**	66. **80**	67. **20**	68. **1**	69. **5**
70. **5**	71. **136**	72. **55**		

Out of syllabus

73. **D**	74. **B**	75. **D**	76. **C**	77. **B**
78. **A**	79. **C**	80. **2**	81. **B**	82. **A**
83. **A**	84. **A**	85. **B**	86. **5**	87. **146**
88. **6952**	89. **D**	90. **C**		

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